PREDICTIVE SALES ANALYTICS ENGINE

Predicting Sales Outcomes from Conversational Intelligence

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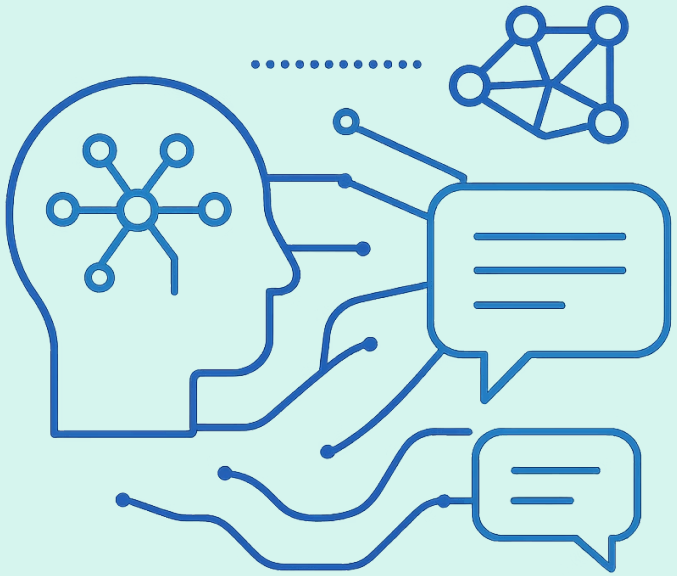

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TASK DEFINITION

Objective & Data Footprint

The core task is predicting deal outcome from raw B2B SaaS sales conversation transcripts — no hand-crafted features, no structured CRM signals. The model must extract signal entirely from language.

100K Conversations

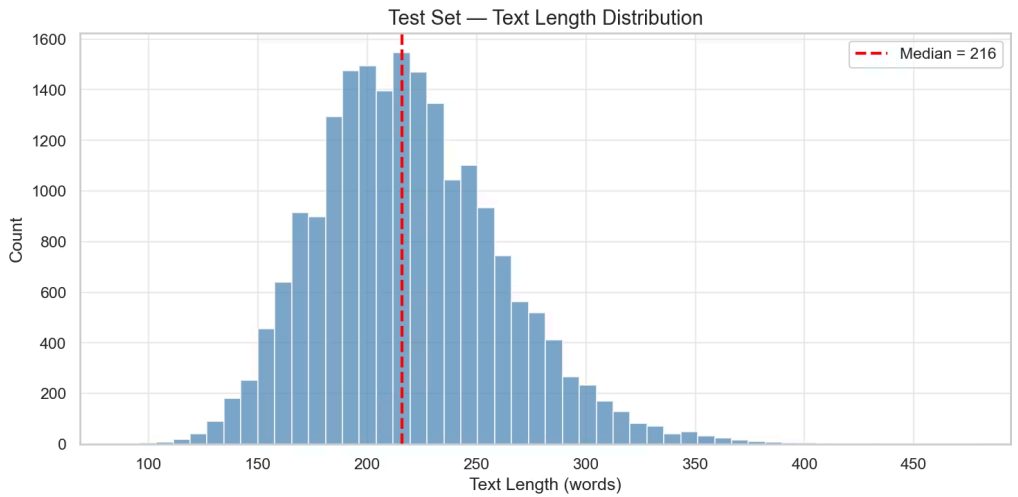
Full training corpus, unfiltered

≤512 Tokens

No truncation required — sequences fit within model capacity

Key Challenge

Long-range dependencies + noisy conversational language



Token length distribution across 100K conversations. The vast majority fall well within the 512-token budget, validating the no-truncation design decision and eliminating a major source of information loss.

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NLP Preprocessing — Custom Tokenization

Design Decision

Custom word-level tokenizer with a 28K vocabulary, built from domain-specific corpus. Preserves SaaS-critical semantic units — **ARR**, **upsell**, **pricing tier**, **churn risk** — that subword models routinely fragment.

Trade-off: Vocab Size vs. Domain Fidelity

Subword tokenizers (BPE, WordPiece) split rare domain terms, degrading downstream representations. A word-level tokenizer with a larger vocab incurs higher embedding matrix cost but preserves token semantics critical for deal-outcome signal.

⚠ Larger vocabulary → higher memory footprint and sparser gradient updates on rare tokens. Mitigated via GloVe initialization (Slide 4).

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Embedding Strategy — Initialization Matters

GloVe initialization yielded +1.11% F1 and +1.27% ROC-AUC over random init — detailed results in Slide 9. The embedding layer was kept trainable to allow domain fine-tuning.

Random Initialization

Embeddings trained from scratch on 100K samples. Sufficient data for frequent tokens, but rare domain terms receive sparse gradient signal and converge slowly.

GloVe 6B — 100d

Pre-trained global co-occurrence statistics provide semantic grounding for rare tokens immediately at epoch 0. Common linguistic patterns transfer effectively, accelerating convergence and improving generalization.

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Architecture — BiLSTM Backbone

hidden_dim = 128 per direction

Balances representational capacity with regularization pressure on a 100K dataset.

Orthogonal Initialization

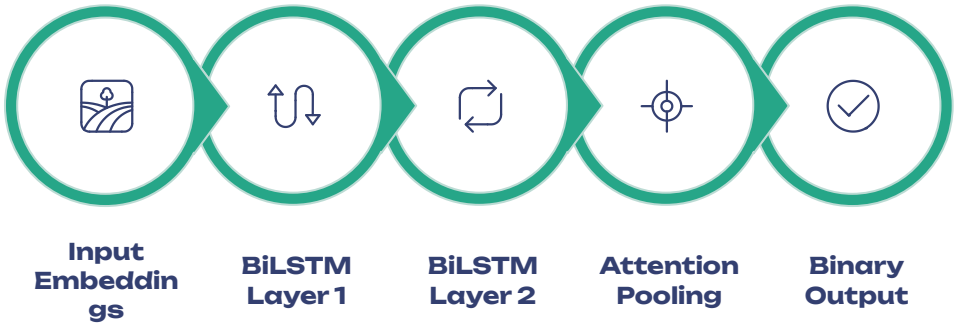
Preserves gradient norm through recurrent steps, mitigating vanishing/exploding gradients on sequences approaching 512 tokens.

Long-Sequence Robustness

LSTM gating (input, forget, output) explicitly manages information flow across long conversational arcs.

Design Rationale

A 2-layer Bidirectional LSTM captures both left-to-right and right-to-left temporal dependencies — critical for sales conversations where closing signals often contextualize earlier objections.



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Attention Mechanism — Bahdanau Self-Attention Pooling

Architecture Note: The attention scorer is a single-layer MLP with tanh activation, followed by softmax normalization. Output is a context vector $\mathbf{c} = \Sigma\, \alpha_i \mathbf{h}_i$, fed directly to the classification head.

Rather than using the final hidden state as sequence representation, Bahdanau-style additive attention computes a scalar score per timestep and produces a weighted sum over all hidden states. This allows the model to selectively attend to high-signal segments regardless of position.

Pricing Objections

High-weight segments correlating with deal loss

Negotiation Cues

Late-stage urgency signals correlating with deal close

Interpretability

Attention weights double as post-hoc saliency maps for debugging

Attention pooling consistently outperformed max-pooling and mean-pooling baselines, confirming the benefit of learned position weighting over fixed aggregation strategies.

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Training Infrastructure & Regularization

Optimizer: AdamW

Decoupled weight decay prevents L2 regularization from interacting with adaptive learning rate scaling. Weight decay = 1e-2.

Scheduler: ReduceLROnPlateau

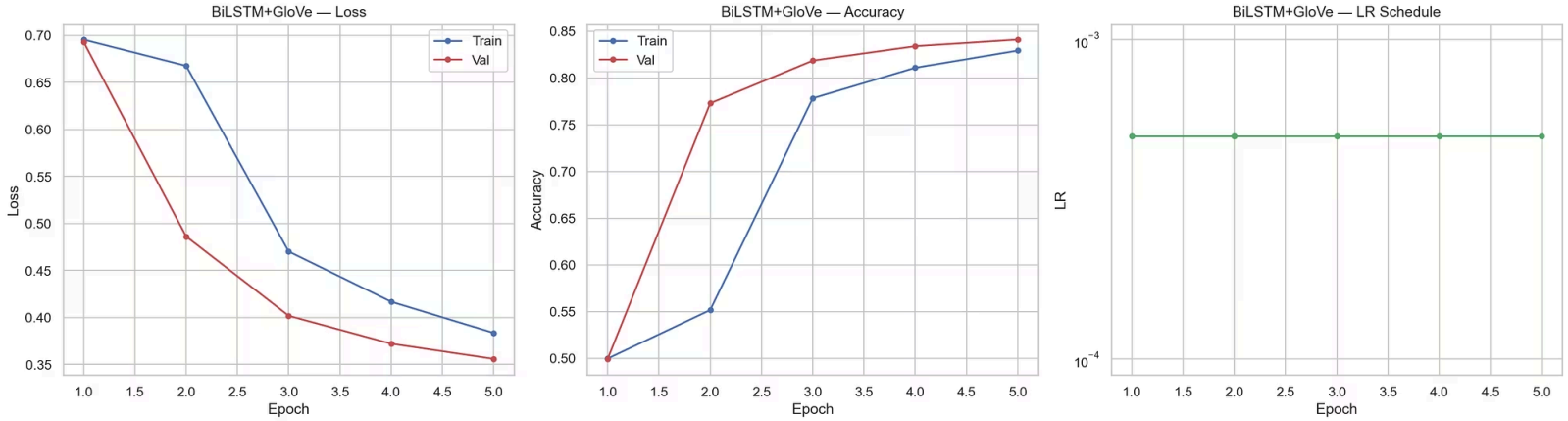
Halves LR on validation loss plateau (patience=3). Prevents stagnation without committing to a fixed annealing schedule.

Dropout (p=0.5) + LayerNorm

Applied post-LSTM and post-attention. LayerNorm stabilizes activations across variable-length sequences.

Gradient Clipping (norm=1.0)

Hard constraint on gradient norm prevents LSTM exploding gradient pathology on long sequences.



Training and validation loss curves for BiLSTM + GloVe. Smooth monotonic convergence with no divergence or overfitting artifacts — confirms the regularization stack is appropriately calibrated for this dataset scale.

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TRANSFORMER TRACK | DISTILBERT CONSTRAINTS

Transformer Track — DistilBERT Constraints

Model: DistilBERT

~66M parameters. 40% smaller than BERT-base via knowledge distillation, retaining ~97% of BERT's performance on GLUE benchmarks. Selected as the transformer track candidate for its inference efficiency.

The Bottleneck: Infrastructure, Not Architecture

Fine-tuning was constrained to **2,000 samples** due to GPU memory limits at batch processing scale. DistilBERT's self-attention is $O(n^2)$ in sequence length — at 512 tokens and large batch sizes, VRAM was saturated before the model could meaningfully generalize.

Root Cause

Memory-bound, not model-bound. Full 100K fine-tuning was architecturally viable but operationally blocked.

Mitigation Path

Gradient checkpointing + mixed-precision (fp16) would recover ~40% VRAM, enabling larger batch sizes and full-corpus training.

Fair Comparison

DistilBERT results are excluded from the primary benchmark — an undertrained transformer is not a representative signal.

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Results — Embedding Optimization

BiLSTM: Random Init vs. GloVe Init

Model	Accuracy	F1	ROC-AUC
BiLSTM (Random)	82.77%	83.33%	0.9102
BiLSTM (GloVe)	**83.89%**	**84.44%**	**0.9229**

+1.11%

F1 Improvement

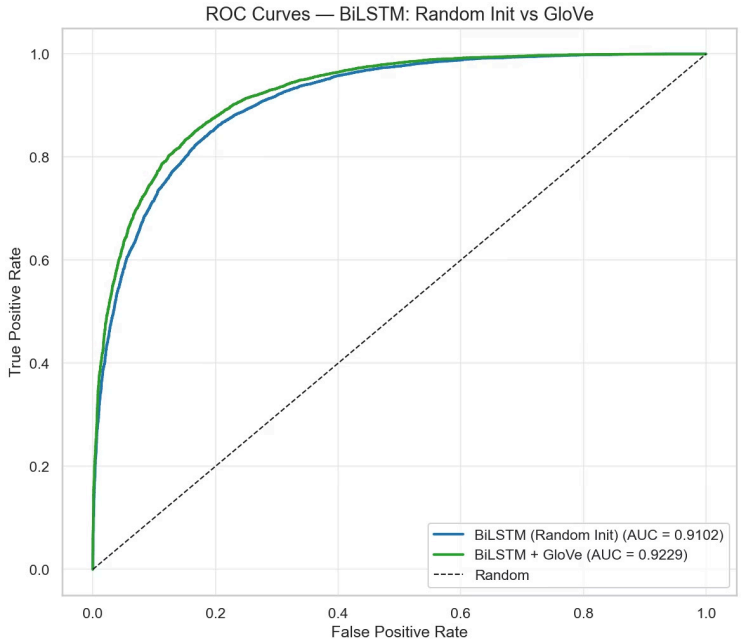
GloVe vs. Random Init

+1.27%

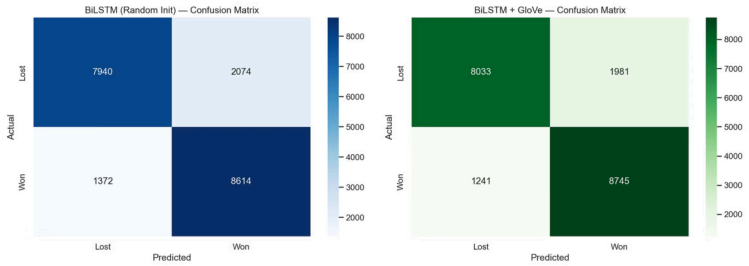
ROC-AUC Gain

Consistent across thresholds

GloVe initialization provides consistent gains across all metrics. The AUC improvement confirms the benefit is not threshold-dependent — the model's ranking quality improves globally.



ROC curves: GloVe init dominates Random init across all operating points. Below: confusion matrices showing reduced false negatives under GloVe initialization — improved classification balance for the minority class.



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Results — DL vs. Classical ML & Error Analysis

Model Comparison

Model	Accuracy	F1
BiLSTM + GloVe	83.89%	84.44%
Logistic Regression	**96.34%**	**96.39%**

Error Analysis

Short Conversations

Low token count → insufficient signal for reliable classification. Accuracy degrades sharply below ~80 tokens.

Ambiguous Negotiation Language

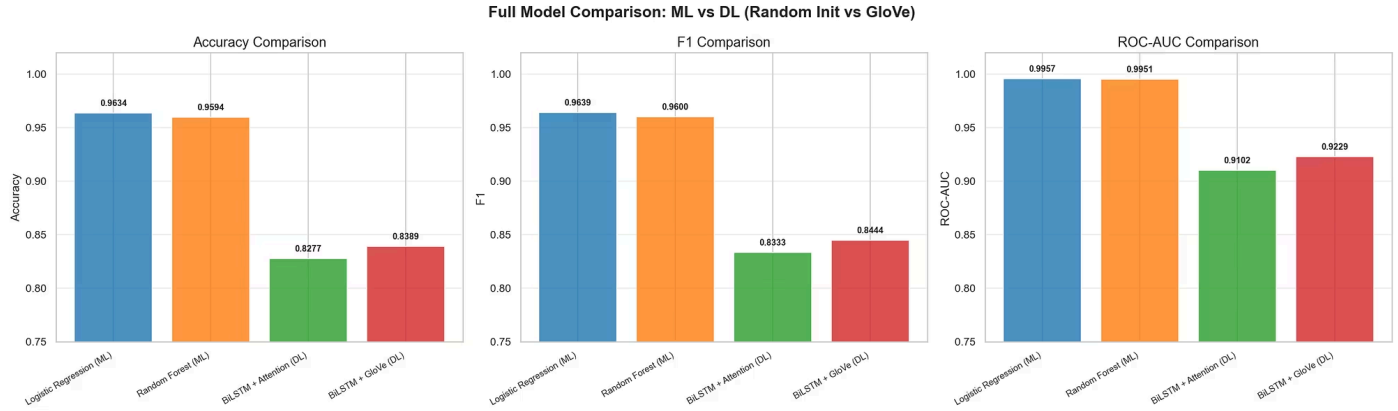
Polite hedging and multi-stakeholder dialogue create noisy mid-confidence predictions that are difficult to separate.

Confidence Distribution

Misclassifications cluster in the 0.4–0.6 confidence range. High-confidence predictions are highly accurate — the model knows when it doesn't know.

Key Insight: Classical ML outperforms due to strong hand-crafted feature engineering. The DL model achieves competitive performance with **zero manual features** — a fundamentally different trade-off between engineering cost and model autonomy.

Accuracy improves monotonically with conversation length — longer transcripts provide richer sequential context for the BiLSTM to exploit. Confidence calibration is well-behaved at the tails.



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